Mafix

Carbon dioxide removal prepurchase application

2025

General Application

(The General Application applies to everyone; all applicants should complete this)

Public section

The content in this section (answers to questions 1(a) - (d)) will be made public on the Frontier GitHub repository after we announce purchases in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Company or organization name

Mafix

Company or organization location (we welcome applicants from anywhere in the world)

Palo Alto

Name(s) of primary point(s) of contact for this application

Jade Marcus and Matt Kanan

Brief company or organization description <20 words

Pioneering the next generation of enhanced rock weathering (ERW) feedstocks that remove CO_2 1000x faster than conventional feedstocks. We are elevating ERW into a more effective, affordable, and readily measurable CDR solution at scale.

1. Public summary of proposed project[1] to Frontier

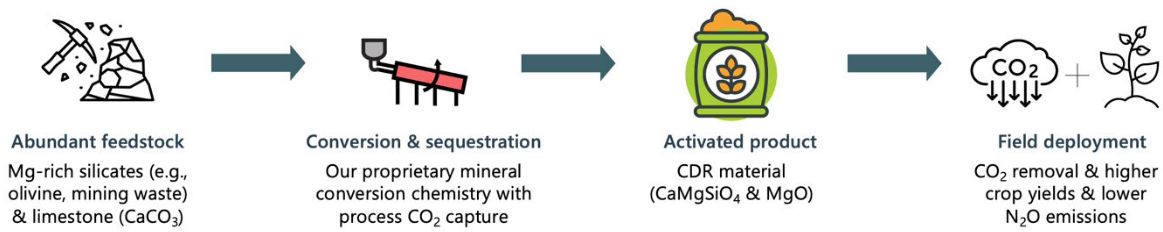
a. **Description of the CDR approach:** Describe how the proposed technology removes CO_2 from the atmosphere, including how the carbon is stored for > 1,000 years. Tell us why your system is best-in-class, and how you're differentiated from any other organization working on a similar approach. If your project addresses any of the priority innovation areas identified in the RFP, tell us how. Please include figures and system schematics and be specific, but concise. 1000-1500 words

How we remove carbon:

Enhanced weathering (EW) approaches to CO_2 removal (CDR) are currently limited by very slow weathering rates of the silicate minerals that are available on Gt scales. To address this barrier, we

[1] We use "project" throughout this template, but the term is not intended to denote a single facility. The "project" being proposed to Frontier could include multiple facilities/locations or potentially all the CDR activities of your company.

have developed a thermal mineral conversion process that transforms limestone ($CaCO_3$) and magnesium silicates into calcium silicates (e.g. $CaMgSiO_4$ known as Monticellite) and magnesium oxide (MgO known as periclase). This mineral composite, which we call "Monti", weathers orders of magnitude faster (up to 1000x faster) than typical EW substrates under ambient conditions relevant for open systems and provides durable removal of CO_2 in the form of stable (bi)carbonate ions. While the mineral conversion process releases CO_2, the amount of CO_2 removed from the atmosphere as (bi)carbonate ions by Monti greatly exceeds the process emissions, enabling deployment at scale using existing assets with no modifications. To further increase the CDR capacity of our technology, we can incorporate CCS in the mineral conversion process and/or can utilize biogenic fuel. By unlocking the alkalinity trapped in the magnesium silicate, we provide incontestable additionality.



When applied to soil, our material can serve as a silicon fertilizer and pH adjuster, providing a substantial increase in plant yield, resistance to pests, and potential reduction in N_2O emissions. Because of the fast weathering of our material, we can achieve these agronomic benefits and measurable CDR at application rates on par with other agronomic inputs (e.g., aglime), which is critical for scalable agronomic adoption unlike existing approaches with basalt and olivine.

How durable is our removal:

When CDR material weathers in soil, the removed CO_2 is ultimately stored as dissolved bicarbonate ions in the ocean, which persist for 10,000–100,000 years. If used in other applications, the removed CO_2 is either stored as dissolved bicarbonate or as precipitated carbonate minerals, which are indefinitely stable.

What differentiates us?

- **Overcome signal to noise issue in MRV**: because of Monti's high weatherability and high alkalinity content, we have been able to get a clear removal signal in every soil we have tested in (>24 different types) unlike existing ERW approaches with noisier data. This greatly simplifies and reduces MRV costs by reducing the number of measurements needed.
- **We are soil agnostic**: we have had measurable CDR in every soil we have tested within 3 months
- **CO_2 capture rate:** we can weather 1000x faster than existing EW substrates that are available on Gt scales.
- **Low energy intensity:** we utilize 60% less energy per ton CO_2 removed when compared to leading direct air capture technologies
- **Cheap feedstocks with multi-Gt/yr abundance:** we can transform *any* Mg silicate into a fast-weathering product. Mg silicates provide an inexhaustible reserve of alkalinity, with >100,000 Gt CDR capacity just from the Mg silicates found in ultramafic rocks.
- **Incontestable additionality:** our innovation unlocks alkalinity presently trapped in silicate minerals, thereby mobilizing this ultra-abundant reserve of alkalinity that would otherwise be unutilized on a relevant timescale.
- **Scalable process & technology:** we can utilize existing rotary kiln assets to produce our materials at scale, one of the only technologies already used for multi-Gt/yr production. By eliminating the need to invent new process tech, we minimize scaling and technology risks.
- **Co-benefits:** Our product offers major agronomic co-benefits as a Si fertilizer. Materials that release silicic acid ($Si(OH)_4$) have well established yield and resilience benefits for crops that are food staples, including rice, maize, soybeans, barley, and tomatoes. Because of its rapid weathering, our product can provide these benefits at low application rates that are feasible for large-scale, economic deployment.
- **Broader GHG impacts:** Beyond the benefits to yield and resistance, our product can be used to tune soil pH to reduce farmland N_2O emissions, which currently account for 50-80% of anthropogenic N_2O emissions.

- **Versatility:** We can tune our material composition and reactivity to meet needs of diverse soils in different geographies. Our product can be deployed for CDR in non-agricultural settings ranging from industrial waste sites to open terrain.

Why us?

We have a uniquely enabling technology: Earth's abundant and accessible alkalinity sources (silicates like basalt, olivine, serpentine) are just not very reactive. Our technology, to our knowledge, is the only product that solves this fundamental problem by converting these abundant feedstocks into completely natural but much more reactive minerals. Our products also have strong additionality as we are unlocking access to new alkalinity, and our cradle-to-grave includes mining, processing, and deployment inefficiencies – ensuring high net CO_2 removal.

We are well positioned for deployment: Along with our partners, we span the value chain from material sourcing to MRV. We have deep expertise across our partners and core members: chemists who pioneered the mineral conversion process, chemical engineers who can scale the process efficiently, agronomists who can optimize the co-benefits, and MRV experts who can quantify removal of CO_2. Our team and partners are excited and energized to be working together to limit warming through carbon dioxide removal by activating our biggest weapon: the rock underfoot.

b. **Project objectives:** What are you trying to build? Discuss location(s) and scale. What is the current cost breakdown, and what needs to happen for your CDR solution to approach Frontier's cost and scale criteria?[2] What is your approach to quantifying the carbon removed? Please include figures and system schematics and be specific, but concise. 1000-1500 words

What are we trying to build

Mafix is pioneering the next generation of ERW feedstocks: fast-weathering, high-alkalinity, all natural materials that weather completely and bypass the signal-to-noise issue that other companies face. We will be the first to deploy and remove all CDR potential in one operating year. With Monti, we can also add value beyond the ERW market by creating a lever for food security through our nutrient release, which we have begun to see in our 12 field trials. We did not find agricultural benefits with olivine and basalt, which were also applied in these trials at the same agronomically relevant application rates.

Our long-term vision is getting to the Gt scale by 2050 without solely relying on the voluntary carbon market (VCM) to do so, but rather on our definitive value as an agronomic input that helps fuel economies worldwide with increased agricultural productivity.

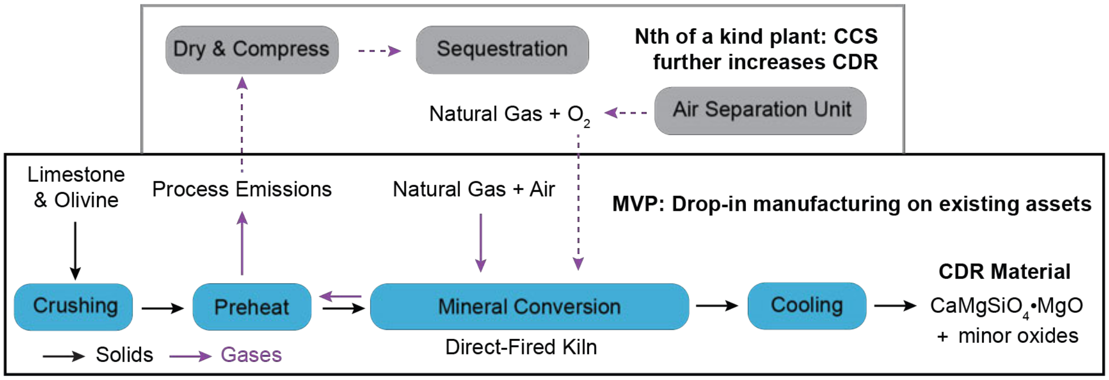
Our proposal to Frontier is to support Mafix as we scale our novel thermochemical process to generate Monti, meaningfully and measurably removing carbon with additionality and durability. We propose partnering on our pilot where we will make Monti in an existing, underutilized cement asset removing ~1,000 net tons CO_2 as our material weathers to form stable dissolved bicarbonate ions. This approach will support us in scaling our production capacity and establishing the agronomic benefits of our material to drive adoption while we onboard key offtakers for our fertilizer. For this project, we will use our proprietary mineral conversion process to produce Monti from limestone and an abundant Mg silicate. We will produce our feedstock in our partner's existing cement facility, thereby de-risking the commercial manufacturing process.

So what is our process?

A high-level schematic of our mineral conversion process is shown in the figure below. The limestone and olivine are first crushed to spec for kiln processing, preheated using the hot flue gas stream, and then fed into a direct-fired kiln, where the limestone is calcined to form CaO and release CO_2, and the CaO then reacts with the olivine to form $CaMgSiO_4$ and MgO, "Monti". The CDR material is cooled

[2] We're looking for approaches that can reach climate-relevant scale (about 0.5 Gt CDR/year at $100/ton). We will consider approaches that don't quite meet this bar if they perform well against our other criteria, can enable the removal of hundreds of millions of tons, are otherwise compelling enough to be part of the global portfolio of climate solutions.

and collected for shipment to distribution sites.



The flue gas from the mineral conversion process contains the CO_2 released from the limestone and an additional smaller amount of CO_2 and H_2O from the kiln combustion. For the Nth of a kind, we will retrofit an existing facility so that the flue gas will be captured and compressed before injection into a pipeline for geologic storage. However, with the MVP, where CO_2 gas is vented, we will still be far more negative than deployments with basalt and olivine on a per ton basis.

Overall, >90% of the energy required to generate our CDR material is thermal energy, which drives down costs and insulates the tech from cost pressures arising from the projected growth in electricity demand in the coming decades. While our FOAK design and TEA/LCA currently assume the use of natural gas as the fuel for mineral conversion, the use of biogas or gasified biomass is a potential carbon-neutral alternative that would further improve the LCA.

How we measure

The CDR material will be transported to our deployment site, leveraging the transportation logistics of our manufacturing partner, and applied at a rate appropriate for the primary land usage. The weathering of the material and resulting CO_2 removed will be monitored and quantified utilizing the protocols of an existing registry, utilizing primary and secondary measurements to quantify removal. Depending on the final application rate, we anticipate complete weathering within a few months to 1 year, which reduces the number of measurements and duration of monitoring needed to verify removal.

How we achieve < $100/ton @ 0.5 Gt/yr

While our initial offer to Frontier is 1,000 tons of CO_2 removal at $500/ton, driven primarily by high MRV and transport costs, we see a clear and achievable path to delivering durable CDR below $100/ton at the 0.5 Gt/year scale. This cost reduction will come from three main levers: logistics optimization, MRV evolution, and value stacking through agronomic benefits.

1. Logistics optimization: Our first deployment centers on a single cement site, but as we expand to additional facilities, we will significantly reduce transport distances and costs. We have established partnerships that allow us to leverage idle kiln capacity within the cement industry. With substantial spare capacity already available, and expected to grow as population growth in G20 nations plateaus, our approach can scale well beyond 0.5 Gt/year by 2050 with minimal or no new capital investment.

2. MRV innovation: As our dataset grows across diverse soils and climates, we can transition from heavy sampling for direct measurement to a model-informed approach. This will dramatically reduce per-ton verification costs while maintaining scientific rigor and transparency.

3. Agronomic value and co-benefits: Our material, Monti, rapidly weathers to release plant-available nutrients (silicic acid, Ca, Mg), improving crop yields and resistance to stress and disease. These agronomic benefits create additional revenue opportunities and help offset CDR costs. Quantifying N_2O reductions will further enhance the climate value of our deployment and align with emerging agricultural regulations targeting these emissions.

Based on current manufacturing, we can produce our CDR material at a cost comparable to clinker production. Given the carbon removal potential per ton, this translates to a pathway below $100/ton CO_2. With over 7 billion acres of global farmland, and more than 1 billion acres expected to benefit directly from our product's nutrient profile, our agricultural deployment alone can achieve 0.5 Gt/year of durable CO_2 removal. Moreover, because our material weathers in all soil types, and even in soil-free conditions when exposed to air and moisture, our addressable market extends well beyond farmland.

c. **Risks:** What are the biggest risks and how will you mitigate those? Include technical, project execution, measurement, reporting and verification (MRV), ecosystem, financial, and any other risks. 500-1000 words

Technical: Our process is more complex than competing EW approaches that rely mostly on grinding and spreading feedstocks, which means that on a rock by rock basis it may look more expensive. However, once you take into account up to 1000x faster weathering (full weatherability in a single year), CDR potential (1.35 ton CO_2 removed per ton CDR material), and simplified MRV because of much higher signal-to-noise, Mafix comes out far ahead.

Importantly, our solution doesn't need to invent any new infrastructure as we can and are utilizing rotary kiln technology that is already operating on the Gt scale. We are leveraging the depth of expertise that is widely utilized today in the cement industry via our partners which allows us to get our product to market immediately.

Financial: As with any engineered CDR technology (DAC, ERW, OAE, etc), there is a risk that buyers will shy away from Gt-scale purchases even at <$100/ton or opt for lower cost but uncertain and impermanent options like afforestation. We believe that the unparalleled quality of the CDR will sustain market growth, but we can nonetheless mitigate this risk by developing the agronomic side of our product as an additional revenue stream. We are currently working with agronomy partners to deploy our materials across a variety of crops and soils to help demonstrate the benefits of Monti as a fertilizer We are gearing up for season 2 across our sites to show YoY benefits!

MRV: Soil is an inherently challenging environment for MRV because of its heterogeneity and high background cation levels. We mitigate the risks of MRV in soil by working across several methods and following the rigorous protocols that have been thoroughly vetted by the scientific community. Critically, we have rigorously demonstrated that Monti weathers across diverse soil types (24 tested so far) through lysimeter & mesocosm experiments that measure exported alkalinity and dissolved inorganic carbon (DIC) in the leachate. To our knowledge, Monti is the only ERW feedstock that has demonstrated substantial weathering and carbon export within the first 3 months of lysimeter measurements. The fast weathering of our material simplifies MRV by reducing the number of measurements required. Eventually, we expect that our material could be deployed with minimal measurements required because its fast-weathering properties will be amply demonstrated across multiple soil types.

Ecosystem: With any new product introduced into an ecosystem, one needs to ensure that there are no unintended, harmful side-effects. While our product is made from only natural inputs (limestone and silicate minerals), there are metal ions present in these feedstocks (e.g. Ni) that are essential at low concentrations but could adversely affect plant health at high levels. We address this risk by using application rates that ensure that impurity metal ions are well within safe limits at 100% weathering. We are doing extensive field trial work and will continue to do this as we deploy our product.

d. **Proposed offer to Frontier:** Please list proposed CDR volume, delivery timeline and price below. If you are selected for a Frontier prepurchase, this table will form the basis of contract discussions.

Proposed CDR over the project lifetime (tons) *(should be net volume after taking into account the uncertainty discount proposed in 5c)*	~1000
Delivery window *(at what point should Frontier consider your contract complete? Should match 2f)*	Before EOY 2027
Levelized cost ($/ton CO_2) *(This is the cost per ton for the project tonnage described above, and should match 6d)*	$302
Levelized price ($/ton CO_2)[3] *(This is the price per ton of your offer to us for the tonnage described above)*	$500

[3] This does not need to exactly match the cost calculated for "This Project" in the TEA spreadsheet (e.g., it's expected to include a margin and reflect reductions from co-product revenue if applicable).